

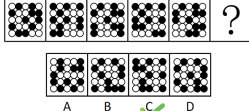
Problem From the four options given, choose the most appropriate one to fill in the question mark so that it shows a certain regularity.  A B C D a	GPT-4o ❌ To determine the correct answer, we need to identify the pattern in the sequence of images. Upon analyzing the sequence, each figure shows a progression where black and white dots alternate positions in a specific and regular way. Observations: 1. In each step, the black dots shift positions, creating a systematic pattern. 2. The white and black dots alternate in a way that spreads uniformly across the grid. By analyzing the progression in the sequence of grids, Option A is the correct choice as it aligns with the observed regularity. e
Reference Solution The elements are the same, so the position pattern is given priority. The figures in the question all have 25 squares, and the number of black blocks in the middle is the same. So the inner and outer circles are considered separately for translation. It is found that the position of the black block in the center of the inner circle remains unchanged, and the other two black blocks move 4 squares clockwise (counterclockwise) each time (it can also be seen as the positions of the two black blocks alternating on the left and right sides), and the outermost black block moves 1 square clockwise each time. Only option C is correct. b	QwenVL2.5 72B-Instruct ❌ To determine the most appropriate option to fill in the question mark, we need to identify the pattern in the sequence of grids. Let's analyze the given grids step by step. The first grid has 10 black circles and 6 white circles. The second grid has 9 black circles and 7 white circles. The third grid has 8 black circles and 8 white circles. We can observe that the number of black circles decreases by 1 and the number of white circles increases by 1 in each subsequent grid. Therefore, the fourth grid should have 7 black circles and 9 white circles. Now, let's check the options: A: This grid has 7 black circles and 9 white circles. B: This grid has 6 black circles and 10 white circles. C: This grid has 8 black circles and 8 white circles. D: This grid has 9 black circles and 7 white circles. The grid that matches our pattern is option A, which has 7 black circles and 9 white circles. Thus, the most appropriate option to fill in the question mark is <code>\boxed{A}</code>. f
InternVL2.5 38B ❌ According to the changing pattern of these graphs, we can observe that: The number of black circles in each graph is gradually decreasing, which allows us to analyze the options in the last row. Let's look at each option: - The number of black circles in options A and B is more than in the previous graphs, which is obviously inconsistent with the pattern. - There are more black circles in option C than before, which is inconsistent with the previous pattern. - The number of black circles in option D is consistent with the previous change pattern. - Therefore, option D is the most suitable option. c	InternVL2.5 38B + RL ✅ To answer this question, we need to carefully observe the sequence of figures and find the pattern between them. First, let's look at the position and changes of the circles in each figure. The number of circles in each figure is the same, but their positions seem to be changing in some way. After careful analysis, we can see that the positions of the circles in the figures are moving in a certain order. 1. In the first figure, the positions of the circles are relatively concentrated. 2. In the second figure, the positions of the circles change slightly and move slightly more than the first figure. 3. In the third figure, the positions of the circles change again and continue to move according to a certain pattern. 4. In the fourth figure, the positions of the circles are closer to the center. 5. In the fifth figure, the positions of the circles continue to move compared to the fourth figure. 6. In the sixth figure, the positions of the circles are different from the other figures, forming a new arrangement. Through the analysis of the above figures, we can see that the current pattern of changes in the figures seems to be gradually forming a new arrangement. Therefore, the seventh figure should continue this change and form a new arrangement. According to this rule, the arrangement of option C is most consistent with the arrangement of the next figure. So, the answer is: C. g
Image Description (GPT-4o) Structure of the Puzzle: • The top row consists of five grids, each with a 6x6 arrangement of black and white circles. • The first four grids follow a specific pattern, while the fifth grid is missing (represented by a "?"). • The goal is to determine the pattern and select the correct option from four answer choices (A, B, C, D) shown below. Observations About the Given Sequence: 1. Each grid in the sequence contains a mix of black and white circles. 2. The arrangement of black circles seems to follow a transformation from one grid to the next. 3. By analyzing the first four grids, a rule or pattern governing the change can be identified. Answer Choices (A, B, C, D): Each of the four answer choices consists of a 6x6 grid similar to the ones in the sequence. Only one of them correctly follows the pattern observed in the first four grids. d	o3-mini ❌ (Reasoned for 5 seconds.) To solve this puzzle, let's follow a step-by-step reasoning approach based on the observations about the sequence of grids. 1. **First Step: Observe the Pattern in the First Four Grids** - We need to closely examine the black and white circles in each grid, paying attention to how the black circles evolve across each grid. - Look at the number of black circles, their positions, and any shifts or transformations from one grid to the next. 2. **Identify Possible Transformations** - The transformation could be based on the movement of black circles, changes in the arrangement (e.g., rotation, mirroring, shifting), or adding/removing black circles in a predictable manner. - Consider changes like: - Moving black circles from one position to another. - Adding or removing black circles. - Symmetry or patterns like shifting rows or columns. 3. **Predict the Missing Grid**- Using the transformations observed from the first four grids, hypothesize what the fifth grid should look like. - Think about the positions, number of black circles, and their movements. The fifth grid should be the natural progression based on the previous grids. 4. **Compare the Answer Choices** - Once you have a hypothesis for the missing grid, compare it against the four answer choices (A, B, C, D). - Choose the one that matches the predicted transformation. Based on this reasoning approach, let's assume that you've carefully analyzed the first four grids and identified a clear pattern in the movement and arrangement of the black circles. Answer: \boxed{B}. h

Figure 6: **Solution examples generated by different models.** Reference solution and outputs generated by GPT-4o [33], Qwen2.5VL-72B-Instruct [8], InternVL2.5-38B [16], and InternVL2.5-38B with RL. Additionally, the image description and solution from LLMs (o3-mini) are also illustrated.

Prompts Setting. We apply three distinct prompting paradigms to investigate model reasoning capabilities: 1) **Non-CoT prompt evaluation:** Models receive a concise instruction: “*Answer the question using a single word or phrase, following this format: Answer: \boxed{\$LETTER}*”. 2) **CoT prompt evaluation:** We prompt models to articulate intermediate reasoning steps: “*Solve the complex visual logical reasoning problem through step-by-step reasoning. Think about the reasoning process first and answer the question following this format: Answer: \boxed{\$LETTER}*”. 3) **Hint prompts evaluation:** Leveraging GPT-4o, we generate question-specific hints derived from the reference solutions. As shown in Figure 7, solution-related hints are provided alongside the CoT prompt to guide reasoning without revealing the final answer directly. Notably, unless otherwise specified, CoT prompt evaluation is employed by default for assessing model performance.

4.2 Overall Results

LLM Performance. Table 1 reports that all evaluated LLMs attain rather low accuracy on VisuLogic. The best-performing LLM, *Qwen2.5-72B-Instruct*, reaches only 28.0%, while *GPT-4* and *Deepseek-R1* achieve 23.6% and 26.6%, respectively. These findings underscore that reasoning based solely on

Table 1: **Cross-Modal performance with CoT prompts on VisuLogic.** The table shows the evaluation scores of baseline references, LLMs, and MLLMs, which illustrates a gap between humans’ and models’ capabilities. Top performers per category are **bolded**, with secondary leaders underlined.

Models	Overall	Quantity	Spatiality	Position	Attribute	Style	Other
References							
Human	51.4	45.3	52.7	71.1	50.0	47.5	44.2
Random	24.9	25.7	25.4	22.7	23.4	24.3	26.1
Open Source LLM (MLLM Description→LLM)							
Deepseek-R1 [18]	<u>26.6</u>	<u>27.7</u>	<u>23.5</u>	24.0	27.8	<u>23.0</u>	35.0
Qwen2.5-72B-Instruct [83]	28.0	30.2	24.4	27.5	<u>26.5</u>	26.8	<u>30.8</u>
QwQ-32B [67]	22.8	24.6	20.1	<u>25.4</u>	19.0	20.7	24.0
Close Source LLM (MLLM Description→LLM)							
GPT-4 (20240613) [1]	23.6	21.2	<u>22.5</u>	21.3	<u>25.6</u>	23.3	35.2
o3-mini (20250131)	24.6	<u>27.8</u>	18.8	24.5	21.7	<u>25.6</u>	28.4
Gemini-2.0-Flash-Thinking (20250121) [64]	23.4	23.2	26.0	16.9	17.1	21.1	<u>33.3</u>
Claude-3.7-Sonnet (20250219)	<u>25.9</u>	26.6	<u>22.5</u>	25.0	28.0	<u>25.6</u>	30.6
Doubao-1.5-Pro-32k (20250115)	26.6	30.0	<u>22.5</u>	25.0	<u>25.6</u>	30.0	24.1
Close Source MLLMs							
GPT-4o-mini (20240718)	24.3	27.2	23.4	23.5	18.3	<u>31.1</u>	16.7
GPT-4o (20240806) [33]	26.3	<u>28.6</u>	24.7	27.2	26.8	20.0	25.9
Kimi-latest [65]	25.9	24.9	29.4	26.5	<u>28.0</u>	16.7	26.9
Doubao-1.5-Vision-Pro-32k (20250115)	28.1	28.1	23.8	29.1	25.1	32.1	35.0
Gemini-2.0-Pro (20250205) [64]	<u>28.0</u>	29.7	24.2	<u>27.9</u>	30.5	22.2	<u>33.3</u>
Claude-3.7-Sonnet (20250219)	24.8	22.7	<u>27.3</u>	<u>27.9</u>	<u>28.0</u>	22.2	22.2
Open Source MLLMs							
LLaVA-v1.5-7B [46]	24.6	26.1	24.2	23.5	17.1	<u>31.1</u>	22.2
LLaVA-OneVision-7B (SI) [39]	25.3	22.4	<u>27.3</u>	33.1	23.2	25.6	22.2
ShareGPT4V [10]	23.4	24.9	22.1	23.5	19.5	28.9	19.4
MiniCPM-o-2.6 [85]	25.3	25.6	23.0	27.3	21.9	24.5	29.9
GLM-4v-9B [25]	24.3	22.4	23.7	28.3	26.0	24.1	25.3
Ovis2-8B [53]	25.6	26.1	23.8	27.2	28.0	25.6	24.1
mPLUG-Owl3-7B-241101 [87]	18.9	21.5	15.2	16.2	20.7	18.9	20.4
Qwen2.5-VL-7B-Instruct [8]	26.0	27.6	20.9	25.2	23.2	37.8	25.0
Qwen2.5VL-72B-Instruct [8]	26.2	25.2	23.8	27.2	25.6	25.6	34.3
QvQ-72B-Preview [66]	23.0	24.2	17.0	24.4	21.0	24.4	30.6
InternVL2.5-38B [14]	25.5	24.4	26.4	27.2	23.2	25.6	26.9
InternVL2.5-78B [14]	<u>27.3</u>	26.6	26.0	26.5	<u>26.8</u>	<u>31.1</u>	30.6
InternVL3-38B [94]	27.1	28.7	27.6	26.1	21.4	23.9	28.5
InternVL3-78B [94]	27.7	<u>27.7</u>	26.1	<u>31.6</u>	26.3	21.3	<u>32.3</u>
Qwen2.5-VL-7B-Instruct-SFT	25.5	24.4	26.4	<u>27.2</u>	23.2	25.6	26.9
Qwen2.5-VL-7B-Instruct-RL	<u>28.0</u>	26.6	33.8	29.4	23.2	18.9	29.6
InternVL2.5-38B-SFT	27.9	<u>30.6</u>	29.4	20.6	<u>25.6</u>	30.0	25.0
InternVL2.5-38B-RL	31.1	31.2	<u>31.2</u>	26.5	30.5	30.0	38.9

textual descriptions is insufficient to capture the rich visual information required by our benchmark, causing failures to resolve visual logical reasoning problems.

MLLM Performance. As shown in Table 1, current multimodal LLMs also perform poorly on VisuLogic. The highest score is 28.1% by *Doubao-1.5-Vision-Pro-32k*, which remains a substantial 23.3 points below human performance. Advanced models such as *GPT-4o* and *Gemini-2.0-Pro* attain only 26.3% and 28.0%, respectively, revealing a marked gap between existing MLLMs and human-level visual reasoning. Overall, these results indicate that current MLLMs have serious deficiencies in visual reasoning and that significant advances are still required.

Effectiveness of CoT Prompts. Contrary to expectations, chain-of-thought (CoT) prompting yields minimal improvements in visual reasoning. As detailed in Table 2, *GPT-4o-mini* benefits most, with only a 1.2-point gain under CoT compared to direct-answer prompts; all other models exhibit gains below 1.0 point. We speculate that this limited effect likely stems from current CoT training being

Table 2: **Influence of Chain-of-Thought on model performance.** Positive value changes are highlighted in **red**, negative changes in **green**, and statistically insignificant variations (delta < 1%) are denoted in **gray**. With CoT prompts, MLLMs only exhibit tiny improvements in visual reasoning.

Models	CoT	Overall	Quantity	Spatiality	Position	Attribute	Style	Other
GPT-4o (20240806)	✓	26.3	28.6	24.7	27.2	26.8	20.0	25.9
	✗	26.0 _(-0.3)	26.9 _(-1.7)	24.2 _(-0.5)	26.5 _(-0.7)	23.2 _(-3.6)	24.0 _(+4.0)	29.6 _(+3.7)
Kimi-latest	✓	25.9	24.9	29.4	26.5	28.0	16.7	26.9
	✗	25.1 _(-0.8)	22.9 _(-2.0)	22.5 _(-6.9)	25.0 _(-1.5)	19.5 _(-7.5)	35.6 _(+18.9)	24.1 _(-2.8)
GPT-4o-mini (20240718)	✓	24.3	27.2	23.4	23.5	18.3	31.1	16.7
	✗	23.1 _(-1.2)	23.8 _(-3.4)	22.9 _(-0.5)	24.3 _(+0.8)	17.1 _(-1.2)	30.0 _(-1.1)	18.5 _(+1.8)
Qwen2.5-VL-Instruct-7B	✓	26.0	27.6	20.9	25.2	23.2	37.8	25.0
	✗	25.9 _(-0.1)	25.5 _(-2.1)	22.8 _(+1.9)	26.4 _(+1.2)	25.3 _(+2.1)	20.6 _(-17.2)	38.2 _(+13.2)
InternVL2.5-38B	✓	24.9	24.1	26.4	27.2	23.2	25.6	22.2
	✗	25.0 _(+0.1)	24.6 _(+0.5)	25.5 _(-0.9)	22.1 _(-5.1)	22.0 _(-1.2)	26.7 _(+1.1)	29.6 _(+7.4)

Table 3: **Influence of hint prompts on model performance.** MLLMs exhibit measurable performance enhancements with hint integration, yet retain significant gaps against human performance. In comparison, humans achieve task mastery on VisuLogic with hints. Value changes are color-coded with **red** indicating positive shifts and **green** denoting negative variations.

Models	Hint	Overall	Quantity	Spatiality	Position	Attribute	Style	Other
Human	✗	51.4	45.3	52.7	71.1	50.0	47.5	44.2
	✓	83.6 _(+32.2)	85.1 _(+39.8)	68.5 _(+15.8)	100.0 _(+28.9)	95.7 _(+45.7)	78.6 _(+31.1)	90.5 _(+46.3)
GPT-4o (20240806)	✗	26.3	28.6	24.7	27.2	26.8	20.0	25.9
	✓	30.0 _(+3.7)	25.4 _(-3.2)	31.5 _(+6.8)	29.2 _(+2.0)	28.6 _(+1.8)	30.8 _(+10.8)	42.9 _(+17.0)
Claude-3.7-Sonnet (20250219)	✗	24.8	22.7	27.3	27.9	28.0	22.2	22.2
	✓	33.5 _(+8.7)	37.3 _(+14.6)	33.3 _(+6.0)	37.5 _(+9.6)	23.8 _(-4.2)	15.4 _(-6.8)	38.1 _(+15.9)
Gemini-2.0-Pro (20250205)	✗	28.0	29.7	24.2	27.9	30.5	22.2	33.3
	✓	36.5 _(+8.5)	44.8 _(+15.1)	33.3 _(+9.1)	25.0 _(-2.9)	38.1 _(+7.6)	15.4 _(-6.8)	42.9 _(+9.6)
Doubao-1.5-Vision-Pro-32k (20250115)	✗	28.1	28.1	23.8	29.1	25.1	32.1	35.0
	✓	37.0 _(+8.9)	46.3 _(+18.2)	25.9 _(+2.1)	54.2 _(+25.1)	33.3 _(+8.2)	23.1 _(-9.0)	28.6 _(-6.4)

based only on pure-text corpora; future works should explore CoT techniques tailored to multimodal data to better support visual reasoning tasks.

Effectiveness of Hint Prompts. Table 3 shows that hint prompts can boost model performance—*Claude-3.7-Sonnet*, *Gemini-2.0-Pro*, and *Doubao-1.5-Vision-Pro-32k* all improve by over 8 points, reaching accuracies above 35%. However, even with explicit guidance, models still fail to construct coherent, reliable reasoning chains. This suggests that simply augmenting training data with similar tasks is insufficient (which can help MLLMs come up with specific directions for solving the problem); future efforts must focus on enhancing the reliability and correctness of reasoning procedures of MLLMs to achieve more accurate reasoning inference.

Impact of Model Scaling. In Table 1, we observe a positive correlation between parameter size and model performance. Within the same model series, *Qwen2.5-VL-72B-Instruct* achieves 26.2 % outperforming *Qwen2.5-VL-7B-Instruct* (26.0%) by 0.2%. Furthermore, *InternVL2.5-78B* (27.3%) surpasses *InternVL2.5-38B* (25.5%) by a margin of 1.8%.

Open-Source vs Close-Source. Table 1 further compares open- and closed-source models. The top open-source MLLM, *InternVL3-78B*, attains 27.7%, trailing the closed-source leader (*Doubao-1.5-Vision-Pro-32k*, 28.1%) by only 0.4% points and outperforming other proprietary competitors such as *GPT-4o* and *Claude-3.7-Sonnet*. Overall, both open- and closed-source models exhibit uniformly low performance, highlighting a widespread neglect of visual reasoning objectives in current multimodal model training and data collection.

Behaviors of RL Trained models. As shown in Table 1, MLLMs with reinforcement learning optimization can yield obvious improvements in visual reasoning performance. *Qwen2.5-VL-7B-Instruct-RL* attains 28.0%, a 2.0 percentage point boost over its non-RL counterpart. More strikingly, *InternVL2.5-38B-RL* reaches 31.1%, surpassing the original non-RL model by 5.6% and establishing a new state-of-the-art on VisuLogic. Furthermore, compared to supervised fine-tuning (SFT) on iden-